In the last 15 years, FPGA engineers have driven market data processing latency from milliseconds down to single digit nanoseconds.

The race is approaching a physical limit.

* Round-trip signal time on co-location cables is less than 100 ns.

NEXT SLIDE

Notion: the next competitive advantage HFT firms can take is smarter execution.

Problem: adding such intelligence to already existing FPGA breaks timing closure.

longer path

Extra logic gates

more jitter

NEXT SLIDE

- A matter of architecture.
- Disadv. of synchronous hardware:
  L> burns power regardless of work being done

transition to → Stalling when moving data from processor to memory.

- Event-driven:
  - Computation fires only as a response to stimuli
  - idle regions consume ≈ 0 W.
  - data moves as small spikes between adjacent cores

This matters because co-location facilities limit firms at a fixed 10 kW per rack.

> The lower energy per inference = more intelligence can fit into that same rack.

[Highlight the Loihi result on the figure. --> Keyword spotting task <--]

NEXT SLIDE

Two independent groups tested SNNs on financial data using an unsupervised learning rule called STDP.

The networks went on to learn temporal patterns in price movements without being told what to look for

—> Both cases show the unsupervised approach yielding higher returns.

—> This result is meaningful for the purpose of this research because it suggests:

L> STDP produces better trading signals than gradient based methods.

L> Native to Neuromorphic chips

Limitations:

L> tests use short time windows.

L> didn't model realistic trading costs.

Generalisability:

L> consistent results across different asset classes

---- NEXT SLIDE ----

- 2 out of 5 actionable gaps

L> POETS gap

L> SNN trading large-scale validation

NEXT SLIDE

The evidence in this survey points to several conclusions.

The one in focus is that the next trading system will be a split pipeline

- FPGAs handle:
  - execution paths
  - order submissions
  - processes that depend on clocked determinism

  ultra-low jitter is a must

- Neuromorphic:
  - pattern recognition
  - signal generation
  - large cross-portfolio risk assessments

  processes where power efficiency determines how much intelligence fits in the co-location rack

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Every component of this architecture has been demonstrated independently in the papers we reviewed. What's missing is an integration of these different architectures.

Building & measuring that split pipeline on a financial workload.